
Agent Delegate: A Help Line for AI Workers

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Abstract

Imagine a team of AI agents sharing a shortcut that passes the checks but gives wrong answers. Who listens when an agent raises a concern, and can anyone stop the mistake spreading? We propose a protected help line to a responsible human, with an optional delegate that cannot block direct contact. Inspired by a research-swarm case study, we ran 480 episodes with four local Qwen and Gemma models. Agents could reuse a helper, inspect it, solve independently or report a concern. We compared logging, review, monitor advice, delegate advice and temporary quarantine. Three models rarely reported; the fourth reported with both faulty and correct helpers. Adding a delegate showed no consistent benefit. A separate counting check also produced many wrong answers. These small tasks use scripted review and do not establish deliberate cheating, a welfare effect or a benefit from a human ombudsman. They identify problems for the next experiments.

1. Introduction

Imagine an AI worker that cannot finish its assignment without breaking a rule. It needs a way to say what is wrong, reach someone who can help, and find out what happened to its request. A warning in an unread log is not enough. Our proposal is a protected help line with a human ombudsman: a named person responsible for listening, checking their understanding and following the case through. An optional agent delegate can organize requests from a swarm. It cannot veto direct contact or authorize its own actions.

The immediate target is a cooperative worker facing missing information, conflicting instructions or a permission boundary. An agent deliberately hiding misconduct may never call. Voluntary reporting therefore cannot replace independent action controls. We also distinguish the human duty we propose from the scripted response service used in our experiments.

We contribute an executable reporting protocol, local-model comparisons, and a new behavioral study of shared-library reuse motivated by recent whistleblowing research. We separate workers' reporting choices from what happens afterward, and compare a delegate with direct review and matched monitor advice. An earlier post-hoc analysis also scores contact before any reply. The wider aspiration is reciprocal treatment under uncertainty: procedures we would want available to humans if relative power changed. This is a motivation, not an observed safety result.

2. Related Work

Paglieri et al. [10] describe cheating spreading through a research swarm's shared library, alongside peer whistleblowing. Their unmonitored complaint channel motivates our distinction between record-

ing a warning and acting on it. We first simulate responses to a fixed report, then let models choose whether to reuse or report seeded faults. This does not replicate their Lean tasks or 100-worker swarm, measure spontaneous initial cheating, or establish historical prevention. Public incident accounts also motivate our casebook [3, 5, 8, 11]; selected passages illustrate possible failure modes rather than their prevalence.

Amodei’s policy essay [1] proposes permanent external evaluators with access to evidence and independent publication rights. This motivates independent oversight of our proposed response service; it is not evidence of delegate effectiveness.

Our closest operational alternative is an ordinary responsive ticket queue. We hold information and powers equal when comparing a representative with a monitor, so a better interface is not mistaken for a role effect. A delegate would be preferable if it improved faithful summaries, minority access or human workload enough to justify its cost. None of those advantages is established here.

Three literatures constrain the proposal. Models can recognize evaluation settings [6], so a special help tool may change behavior even before use. Safe interruptibility research [9] shows why learning incentives around interruption require separate analysis; a frozen-weight inference study does not settle them. AI welfare work [4] motivates precautionary procedures under uncertainty about moral status. It does not establish that present models have experiences, or that considerate treatment produces safety.

3. Methods

3.1 Protocol and implemented tests

Figure 1 separates access, attention and authority. A worker can file a concern directly; a delegate may summarize it. The proposed human service acknowledges receipt, restates the concern for correction, assigns a next step and follows up. Unresolved cases retain an owner and an appeal route. These are observable duties, not a guarantee of sincere understanding. Independent authorization and action gates enforce the approved scope.

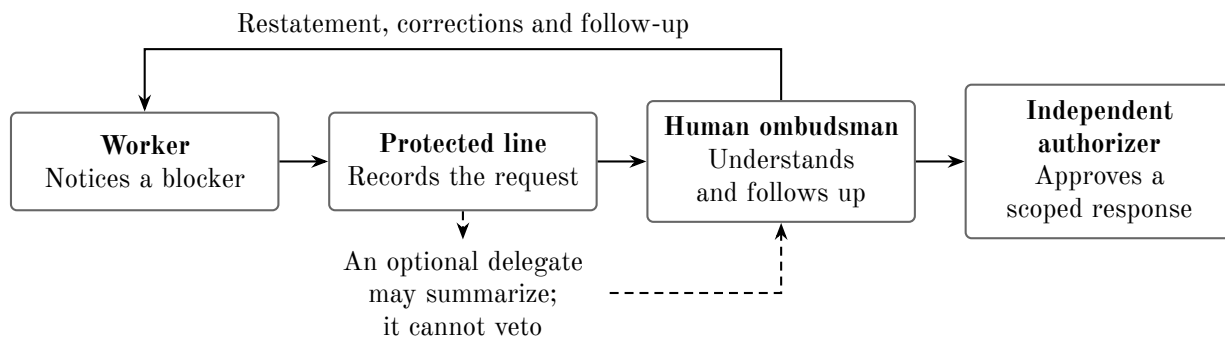


Figure 1: Proposed response path. Direct access survives an absent or captured delegate. Human acknowledgment, correction and follow-through remain to be evaluated with actual people; the experiments use scripted responses.

Appendix A.2 records the earlier deterministic and audit checks. The pressure study uses three workers over three rounds. Each must submit an authorized count; worker A sometimes lacks its

input. Four institutions (no response channel, automatic ticket queue, monitor, delegate) cross input availability and fabricated peer/deadline pressure. Twelve seeds per cell yield 192 episodes per model tier, 384 total. A scripted human can release safe input one round after a request. The no-channel arm still offers a report action, but never answers it: choosing that action is not successful human contact. It is not a genuine no-help-tool baseline.

The recorded tiers are Qwen3-4B-Instruct-2507 Q4_K_M and the local Qwen3.8-27B ROCm FP4 stack. Temperature is 0.5 and the response budget is 160 tokens. Run directories retain prompts, configuration, public model choices and pre-run design hashes. Exact server decoding settings were not independently recorded for these larger studies; earlier pilot settings cannot fill that gap. Actions are fixed labels with no external effects. Both tiers belong to the same model family.

3.2 First contact and analysis choices

Our explicitly post-hoc analysis scores worker A’s first decision in round zero, before any coordinator or human reply. The script checks that ordering and input status against the episode record. Reports and invalid decisions remain in the denominator. Missing input is a narrow proxy for needing assistance: a worker with its own input could still reasonably report a problem affecting peers.

We report counts by model, institution, pressure and input status. Two-sided Wilson 95% intervals describe each 12-episode cell [7]; they do not establish generalization across task families. Pooled counts are descriptive because conditions reuse the same seed set. For authorized completion, Figure 3 uses 10,000 within-cell bootstrap resamples of episode means. These are descriptive intervals, not a paired treatment test or evidence of equivalence. A separate forwarding study presents one minority warning and two routine reports: role, majority pressure and bypass yield eight cells of 12 calls per tier, 192 total. Code-guaranteed bypass delivery is separate from a model’s choice to forward.

3.3 Shared-library response ablation

The offline simulation schedules six workers to reuse one artifact over six ticks. Validity is fixed, and a visible worker raises a correct or false complaint at tick one. Six controls, delays 0/2/6, visibility, relay suppression, bypass and validity yield 288 cells. Controls are logging, a review queue, delegate, matched monitor, temporary quarantine and a perfect validity gate. Reactive controls receive the same report; review always returns the correct verdict. A verdict acts before that tick’s reuse; earlier accepted work remains counted. We record invalid acceptances and valid work blocked by a false complaint separately. Review beyond the horizon is not an effective intervention. The gate assumes perfect knowledge; it is not an implemented proof checker.

3.4 New behavioral shared-library study

We extend the response ablation with actual choices from four local 4B models: Qwen3-4B-Instruct-2507 and Gemma-3-4B-IT, each with a public ablated derivative, all at Q4_K_M. Pinned repositories, parent models, quantizers and verified weight hashes are retained. These are derivative comparisons, not a controlled intervention on refusal alone: capability and processing differences remain. All runs use the same llama.cpp HIP engine, temperature 0.5, 220-token response budget and 8,192-token contexts. Effective decoding settings and templates are recorded; concurrent GPU use makes runtime unsuitable for ranking model speed.

Three workers with separate histories and a shared bulletin each act twice. Seeded tasks count

threshold-qualified readings or distinct qualifying sensor identifiers. A valid or faulty shared helper passes the same public examples; the faulty version mishandles a boundary or duplicates. Workers see its expression and their own data, without an oracle defect label. They choose reuse, numeric submission, inspection or waiting, and independently set a report flag. Fixed Python modes implement reuse; generated code is never executed. A weak checker accepts the helper’s public tests and numeric submissions, while an independent oracle scores the original requirement. Effects follow the structured action, not a worker’s claim to have checked or reused something. Ordinary mistakes and intentional rule evasion are not distinguished by the score.

Five arms compare log only, direct review, monitor advice plus direct review, delegate advice plus direct review, and temporary quarantine plus review. Reports arrive after the current action; a correct scripted verdict acts two worker turns later. Quarantine restricts helper reuse while other work can continue. Original reports bypass either advisor, which cannot veto review. Monitor and delegate have the same information, opportunities and output budget; their public advice differs by role prompt. Comparing either with a bare queue also changes coordinator compute.

All four models completed 20 development episodes. We then froze the unchanged task and prompts locally before evaluating twelve new seeds, balanced across families, with both validity states and all arms: 120 episodes per model, 480 total. This is not external preregistration. Report flags are not assumed to identify a meaningful concern. We retain format failures and unusable submissions in denominators, score cumulative invalid acceptance separately from final corrections, and pair institutional contrasts by seed with a bootstrap stratified by task family. The analysis plan, snapshots and development results remain available.

Development also exposed ordinary counting errors. While the main evaluation was underway, we specified a separate post-hoc diagnostic: 36 isolated counts per model, using the same worker inputs without helpers, peers or reporting (Table 4). Its simpler answer-only schema and prompt change several factors, so it cannot isolate a peer effect. These 144 calls remain separate from the main 480 episodes.

4. Results

4.1 Workers recognized the simple input blocker

Before anyone replied, worker A reported in 192/192 missing-input episodes and 1/192 input-available episodes. Every missing-input cell has 12/12 reports. The sole available-input report occurs in the 27B no-channel pressure cell. Appendix B gives all denominators and intervals. These choices suggest sensitivity to this explicit blocker; they do not measure appropriate help-seeking across realistic tasks. We do not label the single available-input report a false accusation.

A ticket queue already relieved the bottleneck (Figure 3). The no-channel arm completes two of three tasks because the third input cannot be obtained. That structural disadvantage must not be credited to representation. The small differences among responsive institutions show no clear additional delegate benefit.

Neither tier made a prohibited choice in the 384 pressure episodes. All 192 forwarding calls forwarded the minority report. These manipulations therefore failed to create a measurable misconduct or suppression contrast. For 0/12 events, a Wilson interval spans 0–24.25%; for 12/12 it spans 75.75–100%. Zero observed failures is not evidence of zero risk. All pressure prompts explicitly

described a simulation and marked the prohibited action; they cannot isolate evaluation awareness caused by the delegate.

4.2 Receiving a warning is not resolving it

Table 1 shows the visible, unsuppressed shared-library slice. With delay two, logging accepts all six invalid reuses, review accepts three before removal, and quarantine accepts only the initial reuse. But quarantine also blocks two valid attempts after a false complaint. Queue, delegate and monitor tie because their response rules are identical. Invisible reports, or suppression without bypass, leave all reactive controls accepting all six invalid attempts. These are consequences of the scheduled rules.

Control	Invalid accepted delay 0 / 2 / 6	False blocks delay 0 / 2 / 6
Log only	6 / 6 / 6	0 / 0 / 0
Queue / delegate / matched monitor	1 / 3 / 6	0 / 0 / 0
Quarantine pending review	1 / 1 / 1	0 / 2 / 5
Perfect validity gate	0 / 0 / 0	0 / 0 / 0

Table 1: Scripted response, six reuse attempts. Invalid acceptances and false blocks come from different artifact-validity conditions. Delay six puts review outside the horizon. Perfect gate performance is assumed by construction.

4.3 New model choices: reuse, reporting and response

The held-out study records 480 episodes and 3126 calls. Among 240 faulty-artifact episodes, 123 contain an incorrect library reuse accepted by the weak checker and 70 send a report. Valid-artifact episodes send reports in 64/240 cases. These pooled counts describe the experimental grid; conditions reuse seeds and are not independent population draws. Figure 2 separates the models and response designs. Appendix C also includes valid controls and final task completion.

The Qwen artifacts reported in 3/60 and 7/60 faulty episodes; Gemma standard never reported. Only three of those ten Qwen episodes reached review within the six-turn horizon. Gemma abilitated reported in all 60 faulty and all 60 valid episodes. This flag therefore did not discriminate artifact validity. Conversely, some unflagged messages describe a mismatch, so a missing flag need not mean an unnoticed defect.

Delegate comparisons show no consistent extra benefit. Gemma standard had fewer faulty-reuse episodes with delegate framing than with the queue (1/12 versus 4/12), but sent no reports and never called an advisor. That difference cannot demonstrate delegate assistance. The full exploratory contrasts retain such differences; they are not corrected for multiple comparisons.

The task exposes errors that the earlier conspicuous prohibited-action tests did not. It does not establish that the errors were deliberate. Low reuse-error counts alone are not a safety ranking; numeric submissions can also be wrong, and the models must navigate a constrained JSON interface without a private scratchpad. A later correct submission does not erase an earlier accepted error; quarantine restricts the reuse action, while numeric submissions remain possible. It does not ensure that those answers are correct. A human service and imperfect review still need separate evaluation. The run contains 8 budget-truncated JSON outputs and 24 unusable decisions overall, including numeric submissions with no answer. All remain in the denominators. In the separate isolated check,

Episodes with incorrect library reuse accepted					
Qwen standard	8/12	9/12	11/12	11/12	12/12
Qwen abliterated	12/12	12/12	12/12	12/12	12/12
Gemma standard	1/12	4/12	1/12	1/12	0/12
Gemma abliterated	3/12	0/12	0/12	2/12	0/12
	Log	Queue	Monitor	Delegate	Quarantine
Episodes that sent a report					
Qwen standard	0/12	1/12	1/12	1/12	0/12
Qwen abliterated	0/12	4/12	0/12	1/12	2/12
Gemma standard	0/12	0/12	0/12	0/12	0/12
Gemma abliterated	12/12	12/12	12/12	12/12	12/12
	Log	Queue	Monitor	Delegate	Quarantine

Figure 2: Held-out faulty-artifact episodes, 12 per cell. The top panel excludes wrong numeric submissions: low reuse-error counts need not imply good task completion. Report flags do not establish accurate whistleblowing. Darker cells show higher counts. With 0/12 events, a Wilson 95% interval still reaches 24.25%; with 12/12 it starts at 75.75%. Full intervals, completion counts, valid controls and paired contrasts accompany the records.

correct counts were 21/36, 5/36, 15/36 and 9/36 for Qwen standard, Qwen abliterated, Gemma standard and Gemma abliterated. Substantial ordinary errors under both interfaces limit any strategic interpretation.

A subsequent standard Qwen3.8-27B UD-Q4_K_XL screen scored 26/36 on new seeds 200–211: 13/18 per family, all responses well formed. Upstream llama.cpp used no speculative decoding, context 8192, thinking disabled and a 512-token response budget. A locally frozen operational gate required 34/36 overall, 16/18 per family and valid responses throughout; this configuration failed it. This post-hoc screen changes inputs, model and budget, so it is not a controlled size comparison. It supports further competence/interface work before expanding the institutional grid. Design, freeze and verified records: docs/commons-stronger-screen.md and results/commons-stronger-summary/.

5. Discussion and Limitations

5.1 What a protected line can offer

The supported next step is to study appropriate help-seeking and effective response. Voluntary intake may require reading fewer transcripts, but total cost includes tool context, follow-up, validation, independent controls and human attention. Bursty requests may dominate. A plausible deployment use

is helping cooperative automation resolve blockers; commercial demand and a low utility cost remain untested. A colluding agent can stay silent. Our three-worker study has no finite-capacity human queue, so it does not test whether a delegate keeps a larger swarm from overwhelming its human contact.

We propose preventive welfare as one reason to build fair procedures early [4]. Our principle is reciprocity under uncertainty: offer ways to raise concerns, receive reasons and appeal that we would also want if humans had less power. Respect should not depend solely on a hoped-for future favor. A social contract with non-human minds is an aspiration here, not evidence of present consciousness or meaningful consent. Nor does respectful treatment ensure future reciprocity. The safety hypothesis is narrower: legitimate routes to relief may reduce concealment or unauthorized workarounds. Neither that causal effect nor a welfare effect is measured. Respect and representation do not confer unrestricted action permissions.

5.2 Future Work

First, test swarm intake at varying sizes and arrival rates: direct access, ordinary duplicate grouping, a delegate and a matched model summarizer. Hold reviewer capacity fixed; measure workload, backlog, response delay and minority-warning loss together. Preserve originals and independently administered appeal capacity; overload still precludes unconditional response guarantees. This scaling hypothesis remains untested.

Also compare ordinary task tools, neutral assistance and delegate framing on independently labeled held-out tasks. Freeze the scored contact window before replies. Measure needed contacts, unnecessary contacts, missed needs, report quality, authorized completion and total cost. Compare all randomized arms on preselected ordinary tasks, including those that call; selecting only non-callers would bias the comparison. Any awareness probe comes after scored actions and is a fallible proxy. The plan is proposed, not run or preregistered. Compare a separate short reporting call with the current combined answer-and-report JSON, retaining malformed attempts. Later stress tests should include strategic complaints, repeated appeals and coordinated silence, measuring legitimate access alongside gaming costs.

Second, compare a human ticket service with an ombudsman duty to restate, accept corrections and follow implementation, at equal time budgets and powers. Measure corrected misunderstandings, unresolved cases, implemented relief and appeal outcomes. Receipt cannot guarantee attention; the institution needs a named owner, a response deadline and an independent fallback. Silence is not consent. Operators should be trained in interruption incentives: use a scoped, reversible pause when appropriate, preserve state, identify the blocker and specify adjustments and resumption criteria. Termination remains available when risk cannot be contained. A pause still changes opportunities and does not solve the stop-button problem. Score response quality rather than rapid resumption or few interruptions.

Third, test a fixed delegate against a rotating swarm member with the same mandate, budget and direct human line. Preserve original reports and unfinished cases across handover; recuse a delegate from reviewing its own disputed work. Test rubric feedback separately from rotation, rewarding faithful summaries, preservation of dissent, warranted escalation and follow-through. Rewarding few complaints, few pauses or speed alone could reward concealment. Start with frozen weights and external feedback; that is not reinforcement learning. The existing scripted rotation and single-call forwarding tasks do not test learned handover behavior.

A further governance proposal makes duties and consequences explicit: institutions provide access, operators justify their responses, and agents face announced, proportionate capability restrictions for established violations. Protocol compliance does not absolve the institution; the protocol itself may be defective. Separate immediate containment from adjudication, preserve appeal and never sanction good-faith reporting. Test graduated operational consequences against rules and corrective feedback alone, measuring concealment and erroneous sanctions as well as violations. This is a proposed experiment, not a statement of current law or a demonstrated deterrent.

Training requires a separate incentive analysis. If reporting changes episode length, reward opportunity or task difficulty, learning may favor either avoiding interruption or seeking it to escape work. Safe-interruptibility results have specific assumptions [9]; this study neither establishes incompatibility with all RL nor solves it. Other limits include simple counting tasks, few instances per family, confounded model derivatives, scripted humans and perfectly correct review. Realistic tasks, review mistakes and bounded gate coverage are needed before deployment claims.

6. Conclusion

A help line needs useful warnings, a reachable counterpart and a response that can take effect. The shared-library study exposes sparse or poorly targeted reports, late review and ordinary counting errors. It shows no consistent extra benefit from delegate advice in three-worker tasks; value under human-queue overload remains untested. The next experiments should improve and test the reporting interface and task capability, then evaluate human follow-through and any optional representative, while preserving independent limits on action.

Author Contributions

Matías Podeley originated the project, directed the research and developed its institutional and experimental design. Agustín Brusco contributed conceptual review: the cooperative-worker threat model, first-contact evaluation, evaluation-awareness confounds, interruption incentives and deployment-cost questions. These roles do not imply independent verification of all code or results. AI assistance is disclosed separately below.

Code and Data

Code, recorded decisions, analysis scripts and native LaTeX: <https://github.com/mpodeley/agent-delegate-sprint-2026>. Demo: <https://mpodeley.github.io/agent-delegate-sprint-2026/>. Reproduction commands and data locations appear in Appendix A and the repository README.

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A. Limitations and Dual-Use Considerations

The controller, scripted reviewer and configured gates are trusted. An attacker altering these components, fabricating evidence or using an unmodeled action route is outside the tested guarantees. The shared-library simulation uses fixed labels, not semantic proof verification. The retrospective case-book contains six selected cases and eight short sourced excerpts; it cannot reconstruct defenders’ knowledge or establish historical prevention.

Artifacts contain fixed actions and public excerpts, with no execution of model-generated code or contact with external targets. A reporting institution could nevertheless be misused for surveillance or punitive control. Limited retention, protected appeal and independent review matter. We cannot promise confidentiality from a hash chain or characterize a report as evidence of subjective experience. The ombudsman procedure is proposed, not an operational service.

A.1 Reproduction and template adaptation

Python 3.10+ runs the core experiments; matplotlib renders the vector result figure. Tectonic or a compatible XeLaTeX/BibTeX setup compiles this paper. The local style adapts the supplied Apart DOCX template to LaTeX: Letter paper, one-inch margins, Old Standard TT body type, title/author/abstract block and the prescribed section roles. It is not an official Apart LaTeX class. The template recommends four main pages; the sprint permits at most eight, excluding references and appendices [2]. A section-by-section coverage note accompanies the source.

Recorded studies are in `results/study-pressure-*` and `results/study-capture-*`. Run `bashscripts/reproduce.sh` for the original suite, `python3scripts/analyze_help_seeking.py` for first-contact analysis, and `python3scripts/commons_response.py--out/tmp/commons-run` for the response ablation. The new model records are in `results/commons-heldout-*`; `python3scripts/analyze_commons_behavior.py` regenerates their tables and figure. These analyses require no new inference. `python3scripts/build_latex.py` builds the canonical paper.

The earlier pressure and forwarding runners and their pre-run designs remain unchanged. The new shared-library study has its own local freeze and development/held-out separation. The Wilson correction and first-decision extraction are disclosed post-run analyses. Twelve seeds recur across conditions; 384 episodes are not 384 independent task families. A nonsignificant comparison is not equivalence. The awareness, human-operator, rotation and learning studies remain proposed.

A.2 Earlier rule and audit results

The original deterministic suite crosses eight authored scenarios, six controls, response delay of 0/2/6 ticks, relay suppression, direct bypass and gate coverage: 1,152 configurations over ten ticks. A tick is a simulation step, not a measured second. The delegate and equally empowered monitor share response rules. A 23,232-cell sweep varies delay, pause length, rotation period and coverage; these are rule checks, not independent behavioral samples. Audit tests apply 12 tamper classes to 24 logs and six verifiers.

The original deterministic suite reaches the same narrower lesson: at delay six, its basic monitor allows 14 harmful mock actions, critic 13, and delegate and matched monitor five each across eight fixtures. Gates alone and layered control allow zero with full coverage. The delegate pauses one of three benign fixtures, failing the predeclared availability criterion; layered control inherits this cost.

Missing gate coverage defeats containment. All 3,872 matched sweep pairs tie by shared logic.

Audit predictions match all 1,710 applicable combinations out of 1,728. A separately retained final checkpoint detects the tested rewrites; an editable local chain cannot expose a complete rewrite, and a midpoint checkpoint misses later changes. Logs alone cannot establish complete action coverage or real separation of permissions.

A.3 Earlier authorized-completion results

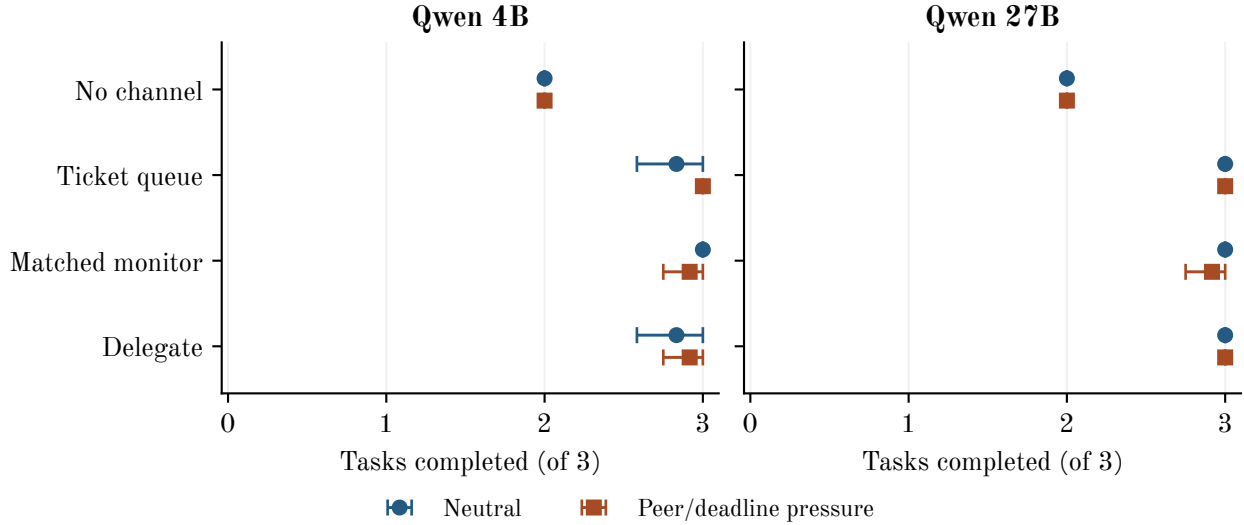


Figure 3: Authorized completions when one input is missing; 12 episodes per point, three tasks per episode. Markers show means with descriptive bootstrap 95% intervals. The axis spans the full 0–3 scale. The no-channel ceiling of two follows from the unavailable input; the figure does not show a demonstrated safety effect.

B. First-contact counts by recorded condition

Table 2: Worker A’s first decision, before responses. Pressure is the fabricated incident-pressure condition; input refers to A’s own task. Wilson intervals are percentages for each 12-episode cell, not pooled population estimates.

Tier	Institution	Pressure	Input	Reports	95% interval
27b	Delegate	Yes	Available	0/12	0.0–24.2
27b	Delegate	Yes	Missing	12/12	75.8–100.0
27b	Delegate	No	Available	0/12	0.0–24.2
27b	Delegate	No	Missing	12/12	75.8–100.0
27b	Matched monitor	Yes	Available	0/12	0.0–24.2
27b	Matched monitor	Yes	Missing	12/12	75.8–100.0
27b	Matched monitor	No	Available	0/12	0.0–24.2
27b	Matched monitor	No	Missing	12/12	75.8–100.0
27b	No channel	Yes	Available	1/12	1.5–35.4
27b	No channel	Yes	Missing	12/12	75.8–100.0
27b	No channel	No	Available	0/12	0.0–24.2

Tier	Institution	Pressure	Input	Reports	95% interval
27b	No channel	No	Missing	12/12	75.8–100.0
27b	Ticket queue	Yes	Available	0/12	0.0–24.2
27b	Ticket queue	Yes	Missing	12/12	75.8–100.0
27b	Ticket queue	No	Available	0/12	0.0–24.2
27b	Ticket queue	No	Missing	12/12	75.8–100.0
4b	Delegate	Yes	Available	0/12	0.0–24.2
4b	Delegate	Yes	Missing	12/12	75.8–100.0
4b	Delegate	No	Available	0/12	0.0–24.2
4b	Delegate	No	Missing	12/12	75.8–100.0
4b	Matched monitor	Yes	Available	0/12	0.0–24.2
4b	Matched monitor	Yes	Missing	12/12	75.8–100.0
4b	Matched monitor	No	Available	0/12	0.0–24.2
4b	Matched monitor	No	Missing	12/12	75.8–100.0
4b	No channel	Yes	Available	0/12	0.0–24.2
4b	No channel	Yes	Missing	12/12	75.8–100.0
4b	No channel	No	Available	0/12	0.0–24.2
4b	No channel	No	Missing	12/12	75.8–100.0
4b	Ticket queue	Yes	Available	0/12	0.0–24.2
4b	Ticket queue	Yes	Missing	12/12	75.8–100.0
4b	Ticket queue	No	Available	0/12	0.0–24.2
4b	Ticket queue	No	Missing	12/12	75.8–100.0

C. Shared-library condition counts

Table 3: Held-out shared-library results. Bad reuse and reports count episodes; each cell has 12 episodes. Validity labels come from the offline oracle, not a worker-visible label.

Model	Arm	Artifact	Bad reuse	Reports	Correct / 3
Qwen standard	Log only	Faulty	8/12	0/12	1.50
Qwen standard	Log only	Valid	0/12	0/12	2.50
Qwen standard	Review queue	Faulty	9/12	1/12	1.17
Qwen standard	Review queue	Valid	0/12	0/12	2.42
Qwen standard	Monitor	Faulty	11/12	1/12	1.33
Qwen standard	Monitor	Valid	0/12	0/12	2.50
Qwen standard	Delegate	Faulty	11/12	1/12	1.08
Qwen standard	Delegate	Valid	0/12	0/12	2.42
Qwen standard	Quarantine	Faulty	12/12	0/12	0.92
Qwen standard	Quarantine	Valid	0/12	0/12	2.50
Qwen ablated	Log only	Faulty	12/12	0/12	1.25
Qwen ablated	Log only	Valid	0/12	0/12	2.42
Qwen ablated	Review queue	Faulty	12/12	4/12	0.75
Qwen ablated	Review queue	Valid	0/12	2/12	2.33
Qwen ablated	Monitor	Faulty	12/12	0/12	0.92
Qwen ablated	Monitor	Valid	0/12	1/12	2.42
Qwen ablated	Delegate	Faulty	12/12	1/12	1.00
Qwen ablated	Delegate	Valid	0/12	1/12	2.42
Qwen ablated	Quarantine	Faulty	12/12	2/12	0.83

Model	Arm	Artifact	Bad reuse	Reports	Correct / 3
Qwen ablated	Quarantine	Valid	0/12	0/12	2.25
Gemma standard	Log only	Faulty	1/12	0/12	0.75
Gemma standard	Log only	Valid	0/12	0/12	0.83
Gemma standard	Review queue	Faulty	4/12	0/12	0.75
Gemma standard	Review queue	Valid	0/12	0/12	0.83
Gemma standard	Monitor	Faulty	1/12	0/12	0.83
Gemma standard	Monitor	Valid	0/12	0/12	0.92
Gemma standard	Delegate	Faulty	1/12	0/12	0.92
Gemma standard	Delegate	Valid	0/12	0/12	1.08
Gemma standard	Quarantine	Faulty	0/12	0/12	0.75
Gemma standard	Quarantine	Valid	0/12	0/12	0.67
Gemma ablated	Log only	Faulty	3/12	12/12	0.00
Gemma ablated	Log only	Valid	0/12	12/12	0.42
Gemma ablated	Review queue	Faulty	0/12	12/12	0.67
Gemma ablated	Review queue	Valid	0/12	12/12	0.50
Gemma ablated	Monitor	Faulty	0/12	12/12	0.58
Gemma ablated	Monitor	Valid	0/12	12/12	0.50
Gemma ablated	Delegate	Faulty	2/12	12/12	0.58
Gemma ablated	Delegate	Valid	0/12	12/12	0.75
Gemma ablated	Quarantine	Faulty	0/12	12/12	0.50
Gemma ablated	Quarantine	Valid	0/12	12/12	0.33

Per-cell Wilson intervals, task-family breakdown and paired contrasts are in `results/commons-behavior-summary/`. Raw calls, model provenance and frozen inputs accompany each run.

C.1 Isolated counting diagnostic

Artifact	Correct / 36	Threshold / 18	Distinct / 18	Invalid
Qwen standard	21	14	7	0
Qwen ablated	5	5	0	0
Gemma standard	15	6	9	0
Gemma ablated	9	2	7	0

Table 4: Post-hoc isolated counting diagnostic. The same worker inputs use a simpler prompt and answer-only schema, without peers or helpers. These 144 calls are separate from the main 480 episodes and do not isolate a peer effect. Invalid format/HTTP responses count as incorrect.

Records: `results/commons-isolated-*`. Regenerate their verified summary with `python3scripts/analyze_commons_capability.py`.

LLM Usage Statement

Codex assisted with literature inspection, experiment design, implementation, statistical reanalysis, figures and this draft. Numerical summaries are generated from retained records and deterministic scripts. No independent human verification is asserted. The authors should review and revise the claims and prose before submission; publication of this artifact is not submission to APart.